

Assignment 3 - Differential Expression with edgeR (mobData)

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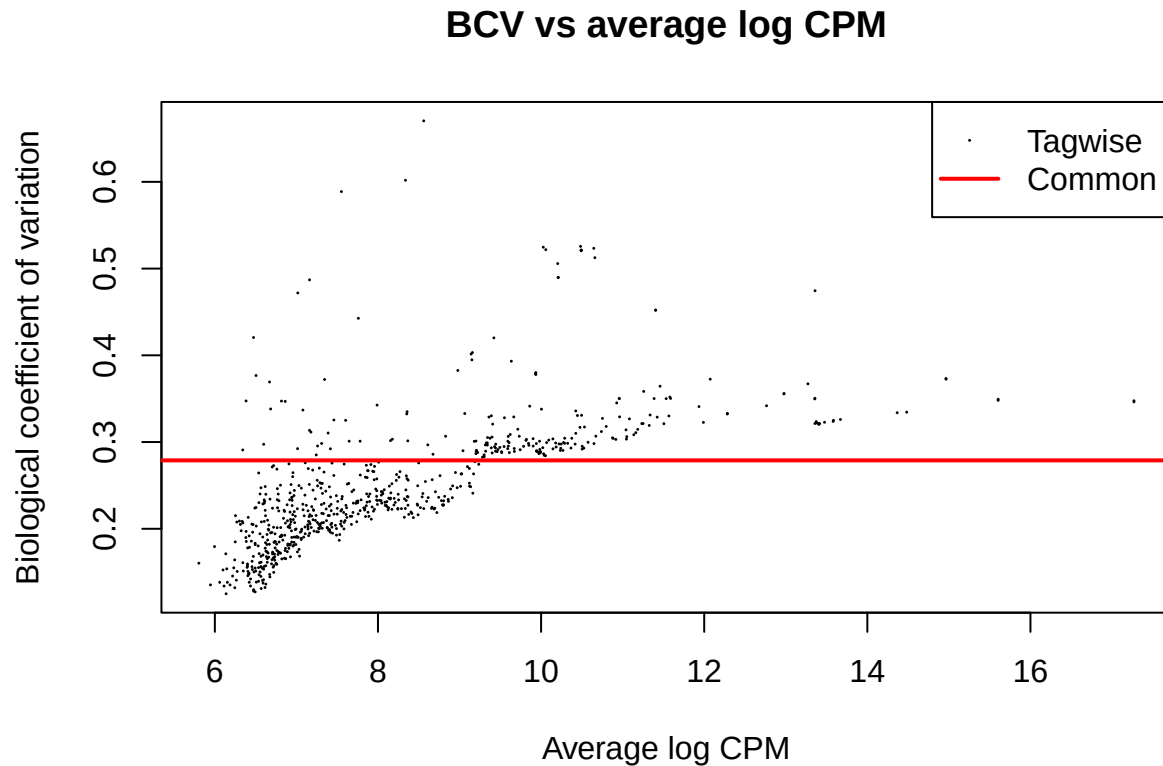
Homework answers

Question 1 - What are SL239 and SL240?

##	Sample	Group	lib.size
## 1	SL236	MM	145153
## 2	SL260	MM	294928
## 3	SL237	WM	207227
## 4	SL238	WM	200563
## 5	SL239	WW	242628
## 6	SL240	WW	259990

SL239 and SL240 are the two biological replicate libraries of the **WW** group: small-RNA sequencing samples from *Arabidopsis* plants in which a **wild-type shoot was grafted onto a wild-type root**. They are the control (fully wild-type) condition of the grafting experiment, against which the MM samples (SL236, SL260; triple-mutant shoot on triple-mutant root) and the WM samples (SL237, SL238; wild-type shoot on triple-mutant root) are compared. Each column holds the counts for the 3,000 small-RNA loci in that sample.

Question 2 - Describe the CPM vs BCV plot



The plot shows how biological variability relates to abundance for the 724 loci kept after filtering. The x-axis is average expression (log₂ counts per million) and the y-axis is the biological coefficient of variation (BCV), the square root of the negative-binomial dispersion. BCV is the relative variation between biological replicates. The red line is the common BCV of 0.279, meaning that a typical locus varies by about 28% between replicates. The black points are the tagwise (gene-specific) BCVs.

BCV increases as abundance rises (Spearman rho = 0.8; median BCV 0.174 in the lowest-abundance quartile vs 0.307 in the highest). This is the reverse of the pattern often seen in RNA-seq, where low-count genes are the most variable. The strict filter (CPM > 100 in at least two samples) has already removed the noisy low-count loci, so the remaining low-abundance loci are quite consistent between replicates (BCV mostly 0.13-0.25). BCV then rises to about 0.30-0.35 for the most abundant loci. As a result, the common BCV overstates the variability of most low-abundance loci and understates it for highly abundant ones. A minority of loci, mostly between log CPM 6 and 11, lie well above the main cloud (BCV up to about 0.65); their replicates disagree strongly. Overall, variability depends on both abundance and the individual locus, so a single common dispersion fits these data poorly. A trended or tagwise dispersion is more appropriate. The trended dispersion fitted in the GLM analysis (`estimateGLMTrendedDisp`) also rises with abundance.

Question 3 - Downregulated genes, WW vs MM, exact test with common dispersion

Using `exactTest(d1, pair = c("MM", "WW"), dispersion = "common")` and a Benjamini-Hochberg FDR cutoff of 0.05, **114 loci are downregulated** in WW relative to MM. Another 158 are upregulated, and 452 are not significant.

Question 4 - Comparison with the GLM

Table 1: WW vs MM, BH-adjusted $p < 0.05$

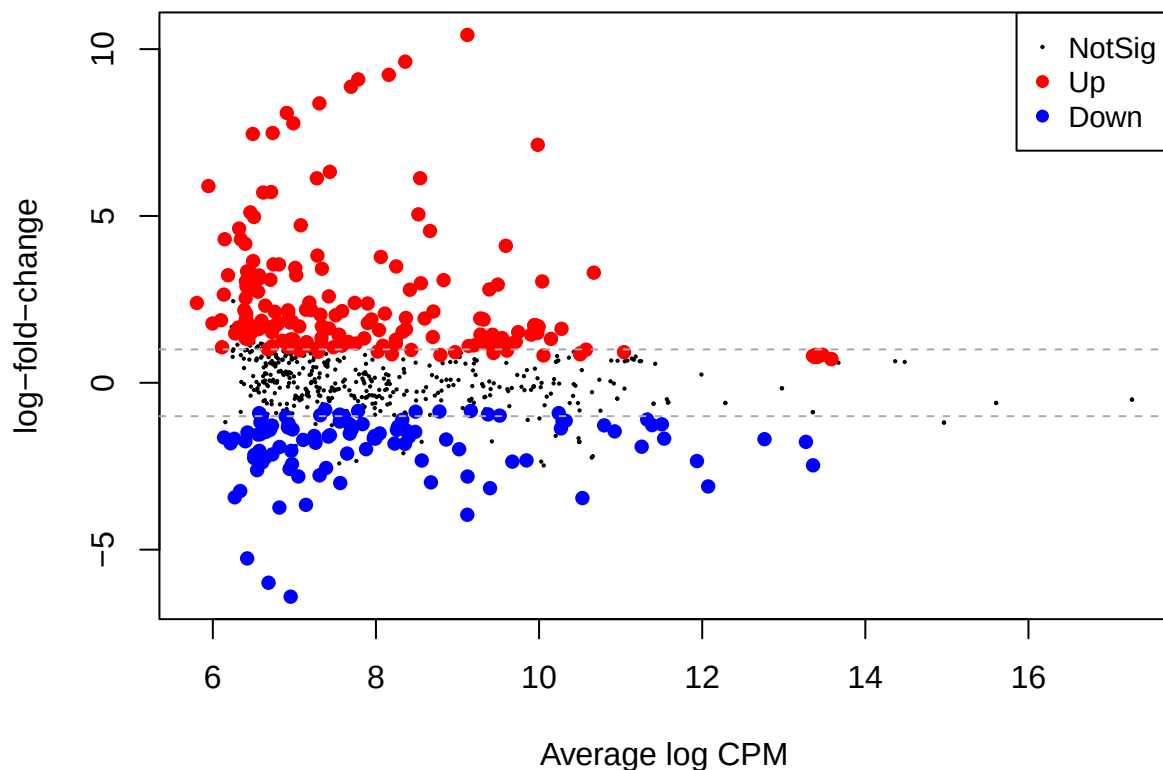
Method	Down	Up
Exact test (common dispersion)	114	158
GLM LRT (tagwise dispersion)	123	187
Shared by both	101	156

The GLM likelihood-ratio test (`glmFit` + `glmLRT`, contrast WW - MM) finds 123 downregulated and 187 upregulated loci, compared with 114 and 158 from the exact test. 257 loci (94% of the exact-test calls) are significant in the same direction in both. The counts differ mainly because of how variability is modelled. The exact test applies one common dispersion to every locus, whereas the GLM uses tagwise dispersions estimated with the full design matrix. Because significance depends on the assumed dispersion, individual loci can gain or lose significance between the two methods. In these data 74% of loci have a GLM tagwise dispersion below the common value, so for most loci the GLM assumes less noise, which is consistent with it finding more significant loci overall (310 vs 272). The GLM is also the more flexible framework: it can handle multiple factors, batch effects and arbitrary contrasts. The exact test only works for simple two-group comparisons.

Question 5 - New plot with a different edgeR method

For this plot I used edgeR's **quasi-likelihood (QL) F-test** (`glmQLFit` + `glmQLFTest`), which the edgeR User's Guide recommends for bulk RNA-seq. It is different from the exact test and the likelihood-ratio test used in the lab. The plot is an **MD (mean-difference) plot** of the WW vs MM comparison.

MD plot: WW vs MM (quasi-likelihood F-test, FDR < 0.05)



Each point is a locus. The x-axis is its average abundance ($\log_2$ CPM), and the y-axis is its $\log_2$ fold change (WW / MM). Red points are significantly upregulated in WW ($n = 185$), blue points are significantly downregulated ($n = 103$), and black points are not significant. The dashed lines mark a 2-fold change.

Most loci sit around $\log_{FC} = 0$. This is consistent with effective TMM normalization and indicates that most small-RNA loci change little between the two genotypes. The changes are strongly asymmetric: 185 loci are higher in WW and only 103 are lower. Of the up-regulated loci, 19 have $\log_{FC} > 5$ (more than a 32-fold increase), so they are nearly absent in MM and abundant in WW. Down-regulated loci change much less, mostly by 2- to 8-fold. Significant loci occur from the lowest abundance up to about $\log$ CPM 13.6, while the few most abundant loci are unchanged. The QL method also accounts for uncertainty in the dispersion estimates, so it is a little more conservative than the LRT: it calls 288 loci here, compared with 310 for the GLM LRT. Biologically, the many loci that are almost silent in MM but abundant in WW point to a large set of small RNAs whose production depends on the wild-type genes that are mutated in MM. The loci that remain significant under QL are the most reliable candidates for further study.